

Exercise 3.09:

a) Begin Read(N) For N = 67, While N < 100 do N = 107 N ← N + 10 EndWhile Write(N) For N = 112, N = 112 End	b) Begin Read(N) For N = 67, N = 107 Repeat N ← N + 10 Until N ≥ 100 Write(N) For N = 112, N = 122 End	Begin Read(N) S ← 0 P ← 1 While N > 0 do S ← S + P * (N mod 2) N ← N div 2 P ← P * 10 EndWhile Write(S) End	For N = 4, S = 100 For N = 12, S = 1100 - What does this pseudo code do?
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Solution of Exercise 3.10 a:

Algo Exo3_10_a Var A, B : Integer Begin Read(A, B) While A ≠ B do If A > B then A ← A - B Else B ← B - A EndIf EndWhile Write(A) End	For A=21 and B=6, A= 3 For A=10 and B=125, A= 5 For A=294 and B=70, A= 14 For A=12 and B=12, A= 12
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Solution of Exercise 3.14_b5:

Algo Exo3_14_b5 Var N, d : Integer Begin Read(N) If N = 0 or N = 1 then Write("Non-prime") Else d ← 2 While N mod d ≠ 0 and d ≤ N div 2 do d ← d + 1 EndWhile If d > N div 2 then Write("Prime") Else Write("Non-prime") EndIf End	<table border="1"> <thead> <tr> <th>N</th><th>Screen</th></tr> </thead> <tbody> <tr> <td>1</td><td>Non-prime</td></tr> <tr> <td>7</td><td>Prime</td></tr> <tr> <td>9</td><td>Non-prime</td></tr> <tr> <td>19</td><td>Prime</td></tr> <tr> <td>49</td><td>Non-prime</td></tr> <tr> <td>1723</td><td>Prime</td></tr> <tr> <td>1729</td><td>Non-prime</td></tr> </tbody> </table>	N	Screen	1	Non-prime	7	Prime	9	Non-prime	19	Prime	49	Non-prime	1723	Prime	1729	Non-prime
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Exercise:

- a- Write an algorithm that requires a natural number and displays the number of its digits.
- b- Write an algorithm that requests a binary number and displays its decimal value.

Home work:

It is requested for the second project to create a program which asks the user to enter four (4) teams from a group of the 1st round of the football world cup, and the results of the 6 corresponding games, in order to display the ranking as following:

```
Please enter the 4 teams:
Uruguay
Republic of Korea
Portugal
Ghana

Please enter the scores:
Uruguay - Republic of Korea : 0 0
Portugal - Ghana : 3 2
Republic of Korea - Ghana : 2 3
Portugal - Uruguay : 2 0
Ghana - Uruguay : 0 2
Republic of Korea - Portugal : 2 1

Ranking
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